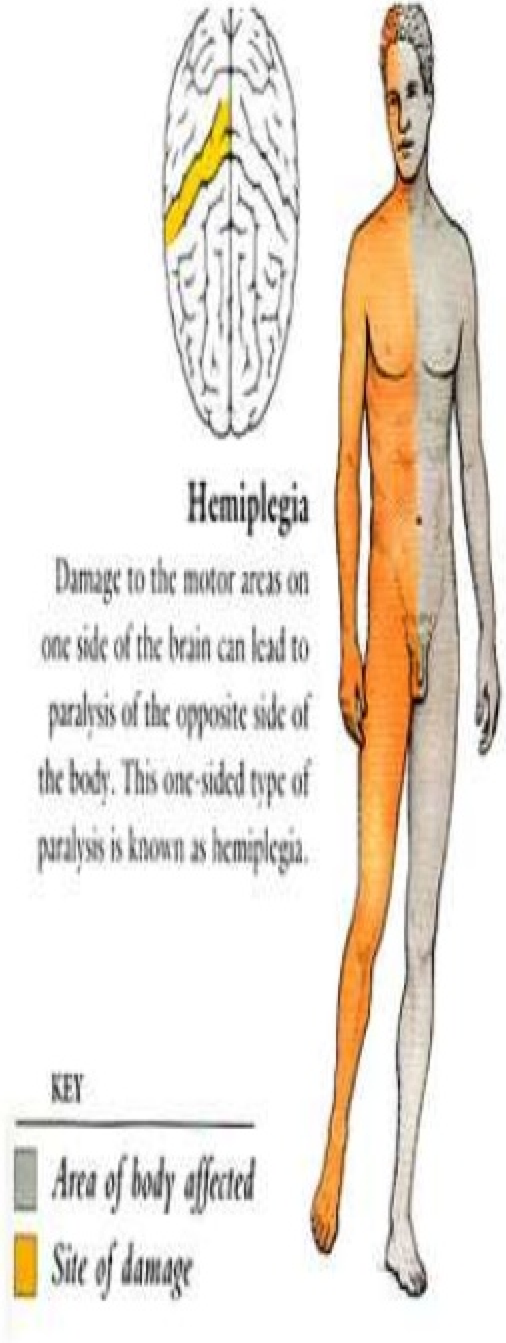
Localization of lesion

- Inability to move a part – Paralysis
- Interruption of motor pathways between cerebral cortex and muscles
- Divided into UMN and LMN
- Both limbs on one side of body affected – Hemiplegia



Features of UMN lesions

- Weakness
- Distal muscle groups affected
- Axial movements, extraocular, upper facial, pharyngeal and jaw muscles are spared
- Hypertonia
- Deep reflex exaggerated
- Loss of abdominal reflex
- Extensor plantar
- No muscle wasting

Complaints of UMN lesion

- Stiffness of legs
- Tendency to trip over
- Unable to walk on rough ground
- Difficulty to climb steps
- Dragging of foot

Features of LMN lesion

- Weakness
- Hypotonia
- Loss of tendon reflex
- Fasciculation of muscle
- Contracture of muscle
- Trophic changes

UMN	LMN
Voluntary activity lost, reflex activity present	Both voluntary and reflex activity lost
Spastic paralysis due to hypertonia, clasp knife rigidity and ankle clonus present	Flaccid paralysis due to hypotonia
Deep reflexes are exaggerated due to increase in muscle tone. Some of the superficial reflexes are lost and some altered	Both superficial and deep reflexes absent
Muscle atrophy occur slowly	Rapid muscular atrophy
Reaction of degeneration absent	Reaction of degeneration present

How to localize UMN lesions

cortex

- Paralysis on opposite side, extent depends on area involved. Unequal weakness of limbs
- Seizure +
- Aphasia + (if dominant hemisphere is involved)
- Loss of cortical sensation
- No movement disorder
- No visual field defect [except in occipital lobe involvement]

Subcortical / Thalamus

- Paralysis on opposite side
- Unequal weakness of limb
- Loss of primary sensation
- Homonymous visual field defect
- No seizure/ no aphasia
- Movement disorder[BG – chorea, parkinsonism, hemibalallismus]

Internal capsule

- Dense hemiplegia and UMN facial palsy

Lesion extending posterior

- Hemianesthesia

- Homonymous hemianopia

Mid brain

- Crossed hemiplegia
- Weber syndrome [3rd nerve palsy]
- Benedicts syndrome [3rd nerve, red nucleus, cerebellar ataxia]

PONS

- Crossed hemipelgia
- Millard Gubler syndrome [6th and 7th nerve palsy]
- Lesion in basis pontis can cause ataxic hemiparesis [weakness + cerebellar on same side]

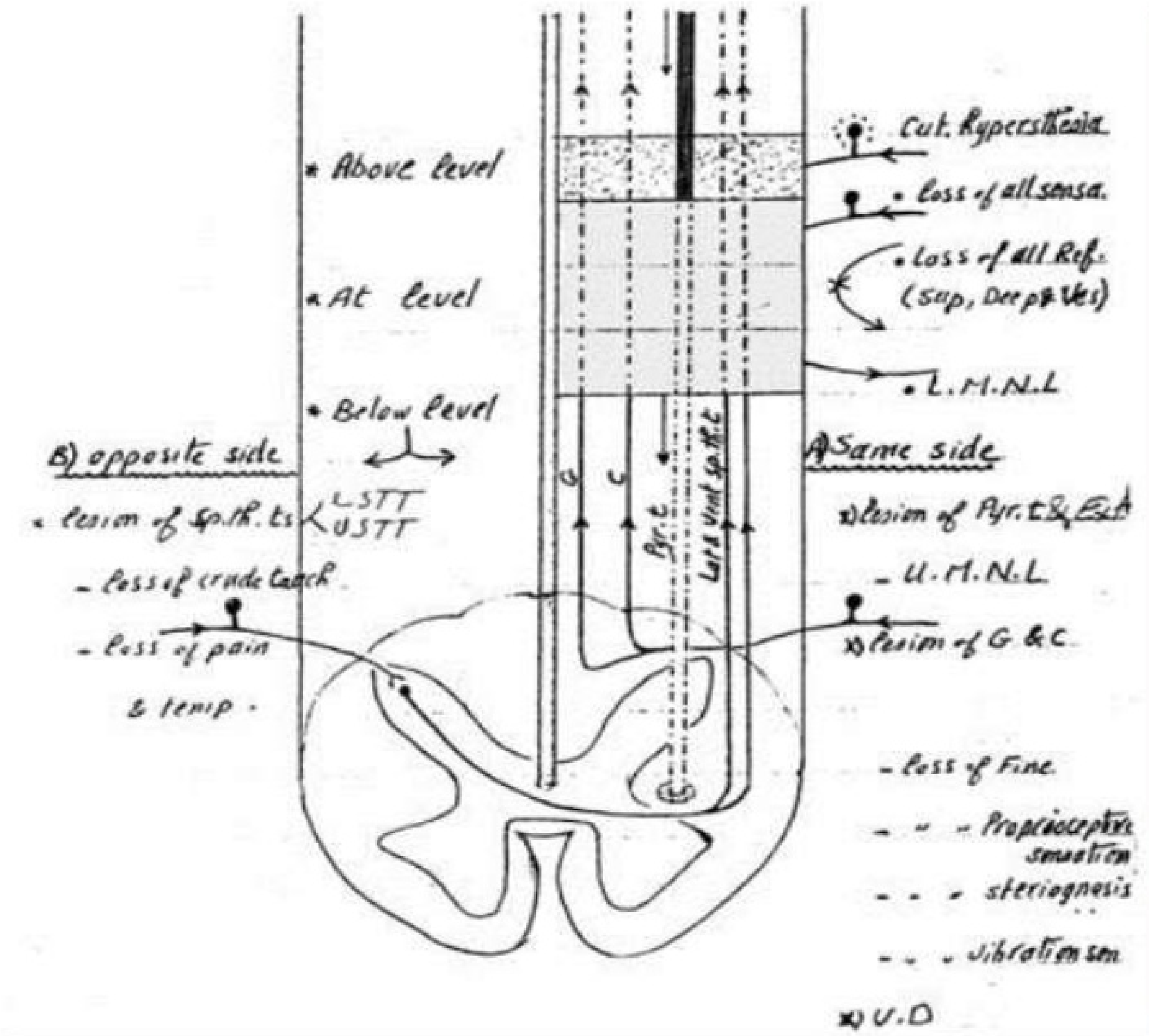
MEDULLA

- Crossed hemiplegia
- Jackson syndrome [12^{th} nerve palsy]
- Involvement of medial lemniscus leads to loss of sensation
- Cruciate hemiplegia

Spinal cord

- Brown sequard syndrome [hemiplegia on the side of involvement without face] lesion above C5
- Effects below the level – UMN type motor loss and sensory loss
- At the level – LMN type
- Above the level – Hyperaesthesia

SPINAL CORD LESION



How to localize LMN lesion

Presenting features	Anterior horn cell disease	Peripheral nerve	Neuromuscular junction disease	Muscle disease
Distribution	Asymmetric limb/ bulbar	Symmetric and distal	Extra ocular, bulbar and proximal limb muscles	Symmetric proximal or distal limb muscles
Muscle atrophy	Marked and early	Moderate	None	Early stage – slight Late – marked
Sensory involvement	None	Parasthesia	None	None
Characteristic features	Fasciculation and cramps	Combined sensory and motor	Diurnal fluctuation	Usually proximal involvement
Reflex	Variable	Decreased	Normal	Decreased

Extra pyramidal system

- Difficulty to initiate movements
- Hypertonia – rigidity [cog wheel]
- No loss of power of muscles and wasting
- Bradykinesia or akinesia
- Involuntary movement
- Plantar response – Flexor

Cerebellar lesion

- Ataxia
- Nystagmus
- Dysarthria
- Intension tremor
- Hypotonia
- Dysdiadocokinesis
- Plantar response – Flexor
- *Paralysis is not a feature of cerebellar disease*